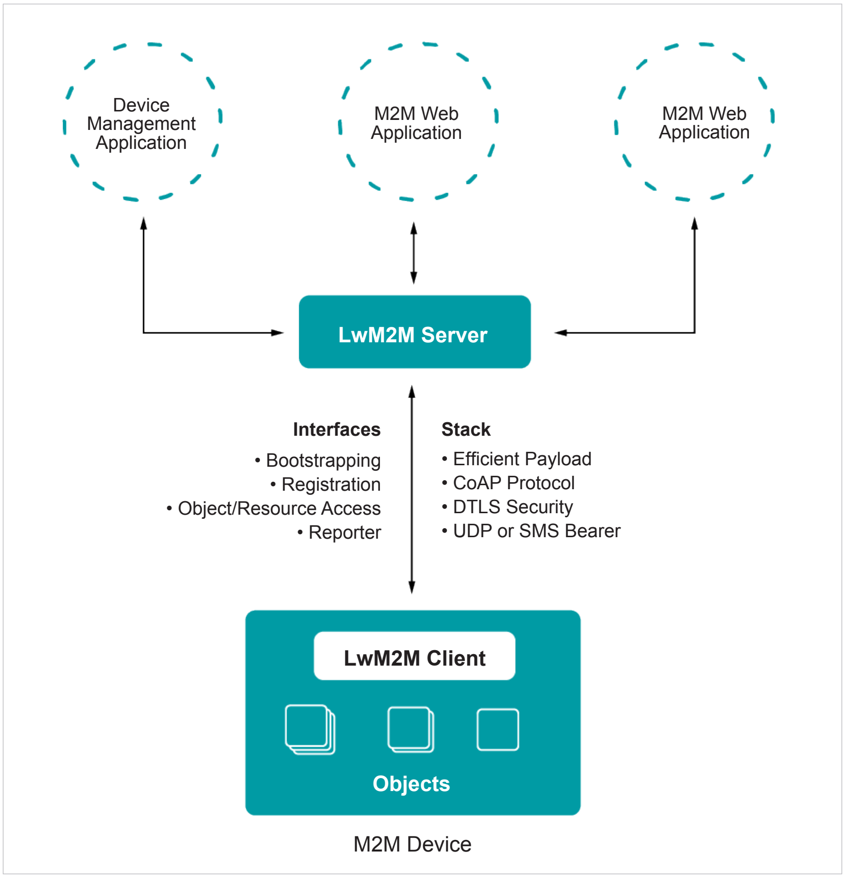

Figure 7: LwM2M data transaction model (Courtesy: AVSystem)

project to project, not only in terms of hardware but also software. For instance, an internal lighting system might require a connection with the cloud while another similar lighting system might require a more localised solution. In some projects, scalability and speed are the top priorities while for others, guaranteed completion of transactions and security might be the focus. As discussed in earlier sections, protocols play a critical role in facilitating such requirements. If the hardware is the body, the protocol is the brain of the device.

Keeping this in mind, I recommend doing a detailed audit and review of the project requirements. It is also important to research industry level IoT projects similar to yours and see which protocol they use. Here's a list of points to consider in this regard.

1. If using a RESTful framework with all the goodness of HTTP is a major requirement for your project, then you cannot go wrong with CoAP. CoAP provides all the features of HTTP, with the additional speed and reliability provided by UDP and message labelling.
2. If your IoT devices have major battery related constraints and you require a lightweight protocol, then MQTT is more suited for your project. It is important to keep in mind that MQTT does suffer with latency issues; so use this protocol only if high latency is not a major concern for your project.
3. If your project is data-intensive and requires fast transaction of messages between multiple devices, then XMPP might be a protocol worth considering. But do this only if the security of the data is not a major concern as XMPP does not have any inbuilt security features. Needless to say, if your project uses XML, then the go-to protocol should be XMPP.
4. If what you are working on are large, high-performance sensor networks such as the ones used in self-driving cars and smart air-traffic control systems, then DDS is probably the ideal choice. However, due diligence is necessary before selecting any protocol, and this is even more true when you are dealing with such advanced projects. DDS is a heavyweight protocol, and requires your devices to have a decent battery life and bandwidth to be able to function.
5. AMQP is a protocol that can be considered for projects that are end-to-end applications — for instance, heavy industrial machinery that has the capacity to handle data-intensive tasks without the concern for power consumption. If transaction completion is an absolute necessity for your project, then AMQP is the most reliable protocol for it.
6. LwM2M is the ideal protocol for telemetry and device management based IoT projects. You can create industry-level large scale IoT projects with high scalability, speed and security. LwM2M has been designed specifically for IoT projects and, as such, has a wide range of applications. You can consider using it for a wide array of projects with varying requirements and consumer bases. END

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